CIFAR-10 Dataset Overview

Objective:

Classify CIFAR-10 images using the k-Nearest Neighbors (kNN) algorithm.

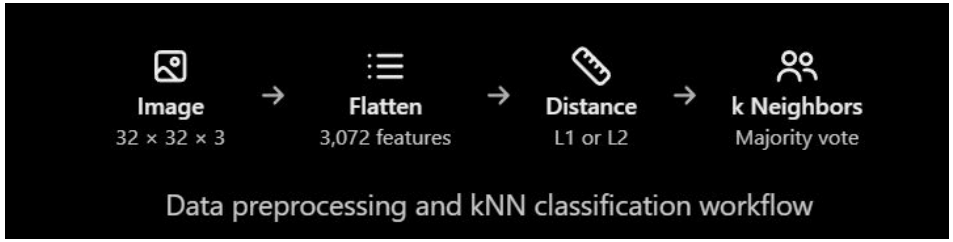
Dataset

- 60,000 RGB images (32 × 32 pixels)
- 10 object categories
- Original split: 50,000 training and 10,000 test images

Experimental Setup

- Training: 5,000 images
- Validation: 1,000 images
- Test: 1,000 images

kNN Methodology



Data Preprocessing

- Flattened each image into 3,072 pixel features.
- Normalized pixel values to the range [0, 1].

Classification

- Calculated distances between images.
- Assigned labels using majority voting among the k nearest neighbors.

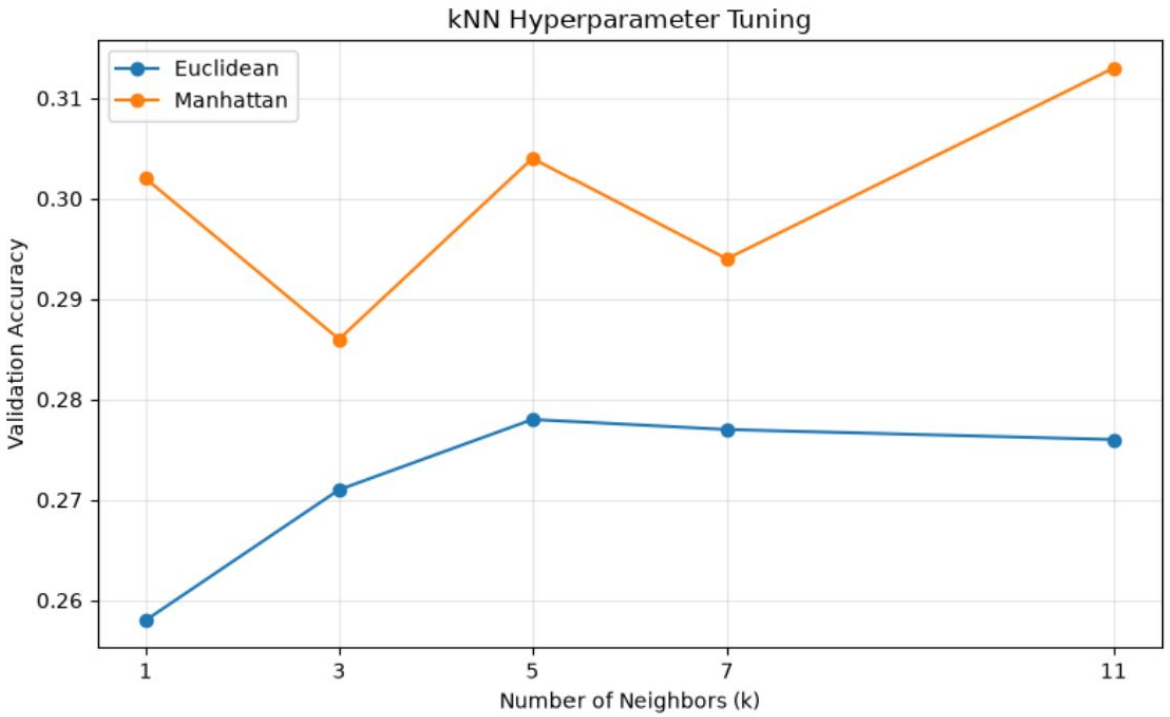
Model Selection

Selected hyperparameters using validation data and evaluated the final model on the test set.

Hyperparameter Tuning

Hyperparameter Tuning

- Tested k = 1, 3, 5, 7, and 11.
- Compared Euclidean (L2) and Manhattan (L1) distances.



Key Findings

- Manhattan distance outperformed Euclidean distance across all tested k values.
- The highest validation accuracy (31.3%) was achieved with k = 11 using Manhattan distance.

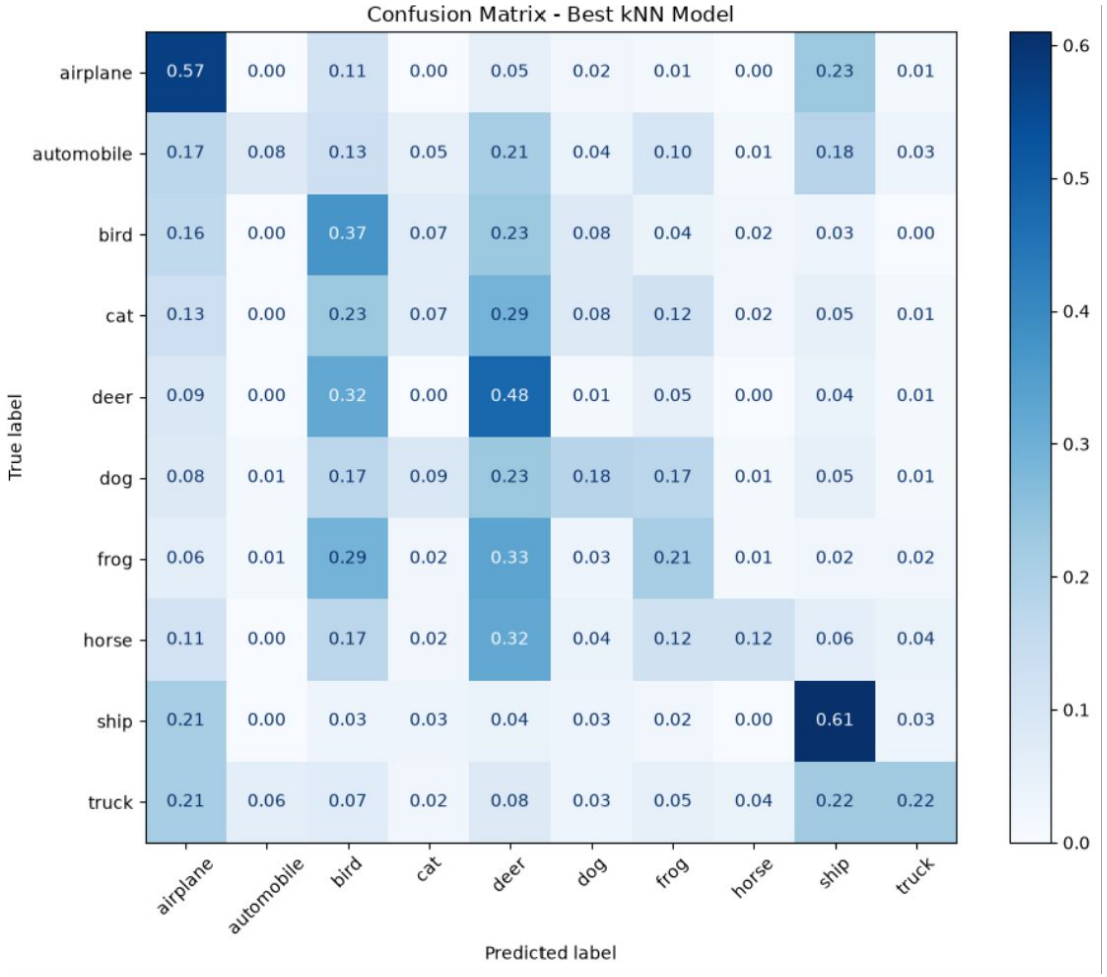
Model Performance

Best Model

- k = 11
- Distance metric: Manhattan (L1)

Test Results

- Accuracy: 29.1%
- Macro F1-score: 0.27



Key Findings

- Ships (61%) and airplanes (57%) showed the highest recall.
- Cats (7%) and automobiles (8%) were particularly difficult to classify.
- The model showed substantial differences in performance across classes.

Failure Cases & Conclusion

Failure Analysis

- The model frequently confused different animal categories.
- Some vehicles were also misclassified as unrelated objects.

Conclusion

- Raw pixel distances may fail to capture meaningful object features.
- kNN provides a simple baseline but showed limited classification performance on CIFAR-10.

